

CSE370 : Database Systems Lab

Database Challenge 02

Activity List

Task 1

Create and use the database “DC_Battles”.

CREATE DATABASE DC_Battles;

Task 2

You have three tables in your database: Heroes, Villains and Battles. All three tables are created for you and the necessary data has been inserted.

.....

```
CREATE TABLE Heroes(  
  Hero_id varchar(3),  
  Name varchar(30),  
  Location varchar(30),  
  Primary key(Hero_id)  
);
```

```
CREATE TABLE Villains(  
  Villain_id varchar(3),  
  Name varchar(30),  
  Location varchar(30),  
  Primary key(Villain_id)  
);
```

```
CREATE TABLE Battles(  
  Battle_id varchar(3),  
  Hero_id varchar(3),  
  Villain_id varchar(3),  
  Battle_Date date,  
  Hero_points int,  
  Villain_points int,  
  Battle_location varchar(30),
```

Primary key(Battle_id),
Foreign key(Hero_id) references Heroes(Hero_id),
Foreign key(Villain_id) references Villains(Villain_id)
);

INSERT INTO Heroes values
('h01', 'The Flash', 'Central City'),
('h02', 'Batman', 'Gotham'),
('h03', 'Green Arrow', 'Star City'),
('h04', 'Wonder Woman', 'Themyscira'),
('h05', 'Green Lantern', 'Coast City'),
('h06', 'Black Canary', 'Star City');

INSERT INTO Villains values
('v01', 'Reverse Flash', 'Central City'),
('v02', 'Deathstroke', 'Star City'),
('v03', 'Joker', 'Gotham'),
('v04', 'Riddler', 'Gotham'),
('v05', 'Harley Quinn', 'Gotham'),
('v06', 'Ares', 'Mount Olympus');

INSERT INTO Battles values
('b01', 'h01', 'v01', '2017-12-12', 85, 90, 'Central City'),
('b02', 'h01', 'v06', '2016-10-09', 40, 98, 'Metropolis'),
('b03', 'h03', 'v02', '2018-06-11', 90, 93, 'Star City'),
('b04', 'h04', 'v06', '2016-10-10', 99, 98, 'Central City'),
('b05', 'h06', 'v05', '2018-07-08', 92, 85, 'Star City'),
('b06', 'h03', 'v02', '2017-06-11', 90, 88, 'Star City');

.....

Please check that all data has been inserted correctly in all three tables.

Task 3

Run appropriate queries to retrieve the requested data below [Use a single query for each question]:

- a. Retrieve the name and average point of each hero.

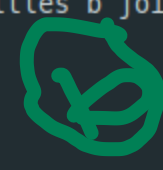
```
MariaDB [dc_battles]> select h.Name, b.Hero_points from Battles b join Heroes h where
```

Name	Hero_points
The Flash	85
The Flash	40
Green Arrow	90
Wonder Woman	99
Black Canary	92
Green Arrow	90

```

SELECT
  H.Name AS Hero_Name,
  V.Name AS Villain_Name,
  B.Battle_Date,
  B.Battle_location
FROM Battles B
JOIN Heroes H ON B.Hero_id = H.Hero_id
JOIN Villains V ON B.Villain_id = V.Villain_id
WHERE B.Hero_points < B.Villain_points;
6 rows in set (0.001 sec)

```



- b. Retrieve the name of heroes, name of villains, date and location of battle where the heroes have scored lower than the villains.

```
MariaDB [dc_battles]> select h.name , v.name, b.Battle_Date, b.Battle_location from
tles b where (h.Hero_id = b.Hero_id) and (v.Villain_id=b.Villain_id) and (b.Hero_po
```

name	name	Battle_Date	Battle_location
The Flash	Reverse Flash	2017-12-12	Central City
The Flash	Ares	2016-10-09	Metropolis
Green Arrow	Deathstroke	2018-06-11	Star City

```

3 rows in set (0.001 sec)

```

- c. Retrieve the name of the villains who have fought the same hero more than once.

```
MariaDB [dc_battles]> select v.name from Battles b join Heroes h on (h.Hero_id = b.H
join Villains v on (v.Villain_id = b.Villain_id) having count(v.name)>1;
```

name
Reverse Flash

```

1 row in set (0.001 sec)

```

- d. Retrieve the name of villains whose locations are not the same as their battle location.

```
MariaDB [dc_battles]> select distinct v.Name from Villains v join Battles b where ((
in_id = b.Villain_id) and (b.Battle_location != v.Location));
+-----+
| Name      |
+-----+
| Ares       |
| Harley Quinn |
+-----+
2 rows in set (0.001 sec)
```

- e. Reverse flash cheated in his battle by time traveling. So he will lose 20% of his points. [Hint: Do not directly use Reverse flash's ID, use subquery]

```
MariaDB [dc_battles]> select (.8*b.Hero_points) as point from Battles b join Villain
(v.Name = 'Reverse Flash') and (v.Villain_id = b.Villain_id);
+-----+
| point |
+-----+
| 68.0 |
+-----+
1 row in set (0.006 sec)
```

- f. Retrieve the name of the hero who achieved the maximum points in all battles.

```
MariaDB [dc_battles]> select h.name from Heroes h join Battles b where (h.Hero_id =
id) having max(b.Hero_points) = (select max(Hero_points) from Battles);
+-----+
| name      |
+-----+
| The Flash |
+-----+
```

- g. Retrieve the name of the villain who achieved the minimum points in all battles.

```
MariaDB [dc_battles]> select v.name from Villains v join Battles b where (v.Villain_
Villain_id) group by v.name having min(b.Villain_points) = (select min(Villain_point
Battles);
+-----+
| name      |
+-----+
| Harley Quinn |
+-----+
1 row in set (0.002 sec)
```

- h. Retrieve the name of villains who have fought more than 1 battle.

f.

```
SELECT H.Name
FROM Heroes H
JOIN Battles B ON H.Hero_id = B.Hero_id
WHERE B.Hero_points = (
    SELECT MAX(Hero_points)
    FROM Battles
);
```

e.

```
SELECT
    B.Battle_id,
    V.Name AS Villain_Name,
    B.Villain_points,
    (B.Villain_points * 0.8) AS
Updated_Villain_Points
FROM Battles B
JOIN Villains V ON B.Villain_id = V.Villain_id
WHERE B.Villain_id = (
    SELECT Villain_id
    FROM Villains
    WHERE Name = 'Reverse Flash'
);
```

```
MariaDB [dc_battles]> select v.name from Villains v join Battles b where (b.Villain_id = v.Villain_id) group by v.name having count(*)>1;
```

name
Ares
Deathstroke

- i. Retrieve the name of the heroes who have fought in more than one distinct city.

```
MariaDB [dc_battles]> select h.name from Heroes h join Battles b on (h.Hero_id = b.Hero_id) group by h.name having (count(b.Battle_location)>1);
```

name
Green Arrow
The Flash

```
2 rows in set (0.001 sec)
```

- j. Retrieve the name of the villains who have the top 3 scores.

```
MariaDB [dc_battles]> select distinct v.Name from Villains v join Battles b on (v.Villain_id = b.Villain_id) order by b.Villain_points desc limit 3;
```

Name
Ares
Deathstroke
Reverse Flash

- k. Retrieve the name of the heroes who have the bottom 3 scores.

```
MariaDB [dc_battles]> select distinct h.Name from Heroes h join Battles b on (h.Hero_id = b.Hero_id) order by b.Hero_points asc limit 3;
```

Name
The Flash
Green Arrow
Black Canary